

RESTFUL BOOKER API TESTING PROJECT

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Application Under Test:

Restful Booker API

Base URL: <https://restful-booker.herokuapp.com>

Application Type:

REST API for Hotel Booking Management

Tools Used:

- Postman
- REST API
- JSON
- HTTP Methods
- API Testing

Testing Scope:

- CRUD Operations Testing
- Authentication Testing
- Request & Response Validation
- Status Code Verification
- Positive Testing
- Negative Testing
- Input Validation Testing

HTTP Methods Covered:

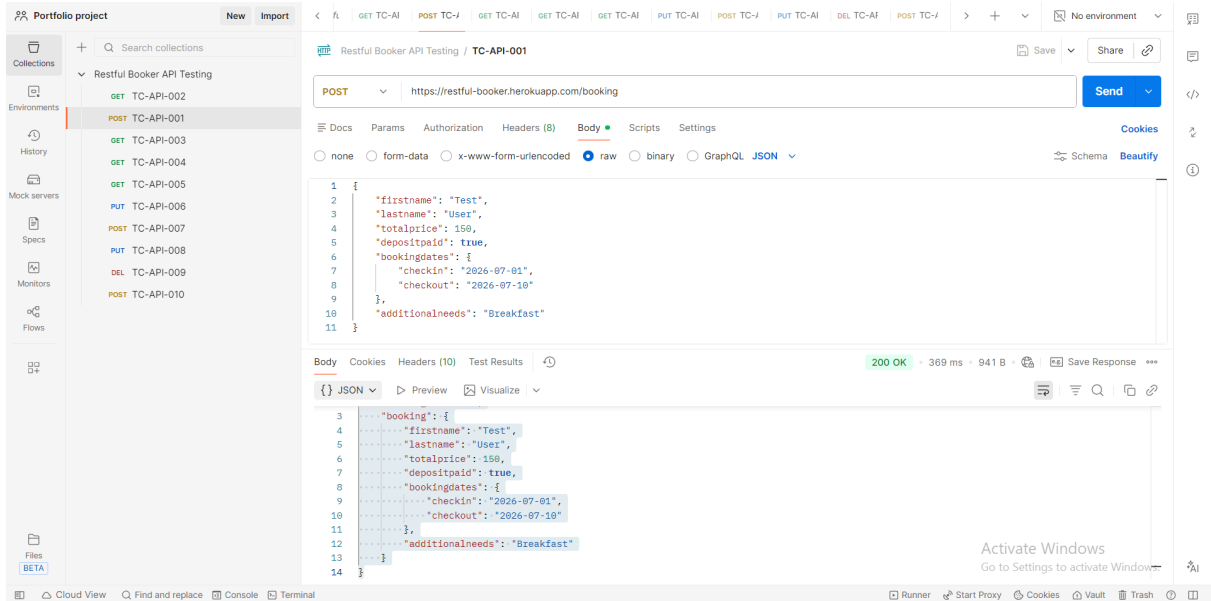
- GET
- POST
- PUT
- DELETE

Artifacts Included:

- 10 API Test Cases
- 1 API Bug Report

Testing Period: June 2026

Postman Collection Structure



TC-API-001

Title: Verify that a booking can be created successfully using valid data

Preconditions:

- * Restful Booker API is available.

Request Method:

POST

Endpoint:

/booking

Request Body:

```
{  
  "firstname": "Test",  
  "lastname": "User",  
  "totalprice": 150,  
  "depositpaid": true,  
  "bookingdates": {  
    "checkin": "2026-07-01",  
    "checkout": "2026-07-10"  
  },  
  "additionalneeds": "Breakfast"  
}
```

Expected Result:

- * Status code 200 is returned.
- * A unique bookingid is generated.
- * Response contains the submitted booking data.
- * Booking details match the request payload.

Priority:

High

TC-API-002

Title: Verify that booking details can be retrieved by booking ID

Preconditions:

- * A booking already exists in the system.

Request Method:

GET

Endpoint:

/booking/{bookingid}

Test Data:

bookingid = 602

Expected Result:

- * Status code 200 is returned.
- * Booking details are returned in the response body.
- * Returned data matches the booking associated with the specified booking ID.
- * Response contains:
 - * firstname
 - * lastname
 - * totalprice
 - * depositpaid
 - * bookingdates

Priority:

High

TC-API-003

Title: Verify that a non-existing booking ID returns 404 Not Found

Preconditions:

- * Restful Booker API is available.

Request Method:

GET

Endpoint:

/booking/{nonExistingBookingId}

Test Data:

nonExistingBookingId = 999999999

Expected Result:

- * Status code 404 is returned.
- * Response body indicates that the booking was not found.

Priority:

Medium

TC-API-004

Title: Verify that all booking IDs can be retrieved successfully

Preconditions:

- * Restful Booker API is available.

Request Method:

GET

Endpoint:

/booking

Expected Result:

- * Status code 200 is returned.
- * Response body contains a list of booking IDs.
- * Response is returned as an array.
- * Each array element contains the field "bookingid".
- * The "bookingid" value is numeric.

Priority:

Medium

TC-API-005

Title: Verify that booking search returns matching results using firstname filter

Preconditions:

* Restful Booker API is available.

Request Method:
GET

Endpoint:
/booking?firstname=John

Expected Result:

- * Status code 200 is returned.
- * Response is returned as an array.
- * The array contains one or more booking IDs.
- * Returned booking IDs correspond to bookings matching the specified firstname filter.

Priority:
Medium

TC-API-006

Title: Verify that booking cannot be updated without authentication

Preconditions:

- * A booking exists in the system.
- * Restful Booker API is available.

Request Method:
PUT

Endpoint:
/booking/602

Test Data:

```
{  
  "firstname": "Updated",  
  "lastname": "User",  
  "totalprice": 200,  
  "depositpaid": false,  
  "bookingdates": {  
    "checkin": "2026-08-01",
```

```
"checkout": "2026-08-10"
},
"additionalneeds": "Dinner"
}
```

Expected Result:

- * The update request is rejected.
- * Status code 403 Forbidden is returned.
- * Booking data remains unchanged.

Priority:
High

TC-API-007

Title: Verify that authentication token can be generated with valid credentials

Preconditions:

- * Restful Booker API is available.

Request Method:
POST

Endpoint:
/auth

Headers:
Content-Type: application/json

Test Data:

```
{
"username": "admin",
"password": "password123"
}
```

Expected Result:

- * Status code 200 is returned.
- * Response contains a token field.

* Token value is generated and is not empty.

Priority:

High

TC-API-008

Title: Verify that a booking can be updated successfully with valid authentication

Preconditions:

* A booking exists in the system.

* A valid authentication token is available.

Request Method:

PUT

Endpoint:

/booking/{bookingid}

Headers:

Content-Type: application/json

Cookie: token=<valid_token>

Test Data:

bookingid: <valid_booking_id>

```
{
  "firstname": "Updated",
  "lastname": "User",
  "totalprice": 200,
  "depositpaid": false,
  "bookingdates": {
    "checkin": "2026-08-01",
    "checkout": "2026-08-10"
  },
  "additionalneeds": "Dinner"
}
```

Expected Result:

* Status code 200 is returned.

* Booking information is updated successfully.

- * Response contains updated booking details.
- * Returned data matches the request payload.

Priority:
High

TC-API-009

Title: Verify that a booking can be deleted successfully with valid authentication

Preconditions:

- * A booking exists in the system.
- * A valid authentication token is available.

Request Method:
DELETE

Endpoint:
/booking/{bookingid}

Headers:
Cookie: token=<valid_token>

Test Data:
bookingid = <valid_booking_id>

Expected Result:

- * Status code 201 Created is returned.
- * Response body contains the message "Created".
- * The booking is deleted successfully.

Priority:
High

TC-API-010

Title: Verify that booking creation fails when First Name is missing

Preconditions:
- Restful Booker API is available.

Request Method:

POST

Endpoint:

/booking

Headers:

Content-Type: application/json

Test Data:

```
{
  "lastname": "User",
  "totalprice": 150,
  "depositpaid": true,
  "bookingdates": {
    "checkin": "2026-07-01",
    "checkout": "2026-07-10"
  },
  "additionalneeds": "Breakfast"
}
```

Expected Result:

- Booking is not created.
- Status code 400 Bad Request is returned.
- Response contains a validation error message indicating that the firstname field is required.

Priority:

High

BUG-API-001

Title:

API returns 500 Internal Server Error when the required "firstname" field is missing during booking creation

Environment:

Postman / Windows 10

API: Restful Booker

Endpoint: <https://restful-booker.herokuapp.com/booking>

Preconditions:

Restful Booker API is available.

Request Method:

POST

Endpoint:

/booking

Headers:

Content-Type: application/json

Request Body:

```
{
  "lastname": "User",
  "totalprice": 150,
  "depositpaid": true,
  "bookingdates": {
    "checkin": "2026-07-01",
    "checkout": "2026-07-10"
  },
  "additionalneeds": "Breakfast"
}
```

Steps to reproduce:

1. Open Postman.
2. Create a POST request to <https://restful-booker.herokuapp.com/booking>
3. Add header: Content-Type: application/json
4. Add request body without the required "firstname" field.
5. Click "Send".
6. Observe the response status code and response body.

Expected Result:

API should reject the request with 400 Bad Request.

Response body should contain a validation error message indicating that the "firstname" field is required.

Booking should not be created.

Actual Result:

API returns 500 Internal Server Error.

Response body: "Internal Server Error"

Severity:

Medium

Priority:
High

Notes:

Missing required fields should be handled by validation logic. Returning 500 Internal Server Error may indicate that the API does not handle invalid input properly.

The screenshot shows a REST client interface with the following details:

- URL:** `https://restful-booker.herokuapp.com/booking`
- Method:** `POST`
- Body:** A JSON object with the following structure:

```
1 {
2   "lastname": "User",
3   "totalprice": 150,
4   "depositpaid": true,
5   "bookingdates": {
6     "checkin": "2026-07-01",
7     "checkout": "2026-07-10"
8   },
9   "additionalneeds": "Breakfast"
10 }
```
- Response:** A `500 Internal Server Error` with a status of `81 ms` and a size of `773 B`. The response body is `Internal Server Error`.